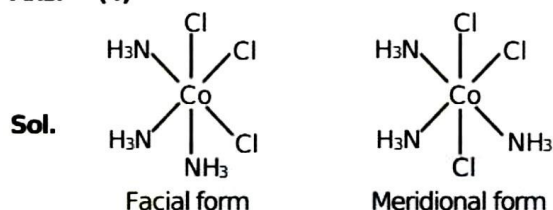


PART : CHEMISTRY

- 1.** Which complex will show facial and meridional form-
 (1) $[\text{Co}(\text{NH}_3)_6]$ (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]$ (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (4) $[\text{Co}(\text{NH}_3)\text{Cl}_3]$

Ans. (4)



- 2.** The depression in freezing point of 0.1 molal solution is 0.558, then complex will be-
 (1) $[\text{Co}(\text{NH}_3)_6\text{Cl}_2]$ (2) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}$ (3) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$ (4) $[\text{Co}(\text{NH}_3)\text{Cl}_3]$

Ans. (1)

Sol. $\Delta T_f = i \cdot K_f \cdot m$
 $0.558 = i \times 1.86 \times 0.1$
 $i = \frac{0.558}{1.86 \times 0.1} = 3$

- 3.** Which of the following element does not lie in same period.
 (1) Osmium (2) Iridium (3) Palladium (4) Platinum

Ans. (3)

Sol.

- 4.** Which of the following pair of ions are same coloured ?
 (1) Ti^{4+} , V^{3+} , Sc^{3+} (2) Cr^{2+} , Cu^{2+} , V^{4+} (3) Cr^{3+} , Ni^{2+} , V^{4+} (4) Mn^{3+} , Fe^{2+} , Zn^{2+}

Ans. (2)

Sol. Cr^{2+} , Cu^{2+} , V^{4+} (Blue)

- 5.** Find ΔG of reaction at 298 K
 $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2\text{NO}_2$;
 $\Delta H = +50 \text{ kJ/mol}$ & $\Delta S = 5 \text{ J/mol}\cdot\text{K}$

Ans. (48.5)

Sol. $\therefore \Delta G = \Delta H - T\Delta S$
 $= 50 \text{ kJ/mol} - 298 \times \frac{5}{1000} \text{ kJ}$
 $= 50 - 1.5 \text{ kJ/mol}$
 $= 48.5 \text{ kJ/mol}$

- 6.** The pH of 0.1 M $\text{C}_2\text{H}_5\text{NH}_2$ solution is 9. if $K_b = 10^{-x}$ then find x.

Ans. (9)

Sol. $\text{pOH} = \frac{1}{2} (\text{p}K_b - \log C)$
 $5 = \frac{1}{2} (\text{p}K_b - \log 10^{-1})$
 $10 - 1 = \text{p}K_b = 9$
 $K_b = 10^{-9} \Rightarrow x = 9$

7. Match the column-I and column-II

Column-I	Column-II
(A) Octet complete	(i) BCl_3 , BeCl_2
(B) Octet expanded	(ii) NO_2 , NO
(C) Octet incomplete	(iii) CCl_4 , CO_2
(D) Odd electron	(iv) H_2SO_4 , PCl_5

- (1) (A) $\rightarrow$ (iii) ; (B) $\rightarrow$ (iv) ; (C) $\rightarrow$ (i) ; (D) $\rightarrow$ (ii) (2) (A) $\rightarrow$ (iii) ; (B) $\rightarrow$ (i) ; (C) $\rightarrow$ (iv) ; (D) $\rightarrow$ (ii)
 (3) (A) $\rightarrow$ (iv) ; (B) $\rightarrow$ (i) ; (C) $\rightarrow$ (ii) ; (D) $\rightarrow$ (iii) (4) (A) $\rightarrow$ (iv) ; (B) $\rightarrow$ (ii) ; (C) $\rightarrow$ (iii) ; (D) $\rightarrow$ (i)

Ans. (1)

8. If 10^{21} molecules are removed from x mg of $\text{CO}_2(\text{g})$ then 2.8×10^{-3} mole are left. Calculate the value of x.

Ans. (196.53)

Sol. $(\text{mole})_i = \left(\frac{x \times 10^{-3}}{44} \right)$, $(\text{mole})_{\text{Removed}} = \left(\frac{10^{21}}{6 \times 10^{23}} \right) = \frac{1}{6} \times 10^{-2}$

$(\text{mole})_{\text{left}} = 2.8 \times 10^{-3}$

Now,

$(\text{mole})_i - (\text{mole})_{\text{Removed}} = (\text{mole})_{\text{left}}$

$= \frac{x \times 10^{-3}}{44} - \frac{10^{21}}{6 \times 10^{23}} = 2.8 \times 10^{-3}$

$\frac{x \times 10^{-3}}{44} = 2.8 \times 10^{-3} + \frac{1}{6} \times 10^{-2} = \left(2.8 + \frac{10}{6} \right) \times 10^{-3}$

$\frac{x \times 10^{-3}}{44} = \left(\frac{16.8 + 10}{6} \right) \times 10^{-3}$

$x = 196.53$

9. Incorrect statement among the following is :

- (1) SO_2 act as oxidising agent but not reducing agent.
 (2) NO_2 exist as dimer
 (3) PF_5 exist but NF_5 does not
 (4) PH_3 has lower proton affinity than NH_3

Ans. (1)

10. Two radioactive decays are



Find ratio of $(N_A)_t$ and $(N_B)_t$ after one half life of A

Ans. (4)

Sol. Radioactive decays obeys 1st order kinetics

$\frac{(N_B)_t}{(N_A)_t} = \frac{N_{B_0} e^{-\lambda_2(t_{1/2})_A}}{N_{A_0} e^{-\lambda_1(t_{1/2})_A}} \quad (N_{A_0} = N_{B_0})$

$= \frac{e^{-\lambda_2 \times \frac{\ln 2}{\lambda_1}}}{e^{-\lambda_1 \times \frac{\ln 2}{\lambda_1}}} = \frac{e^{-\ln 2}}{e^{-\ln 2}} \Rightarrow \frac{(N_A)_t}{(N_B)_t} = 4$

11. Calculate the percentage by weight of S if 160 g of organic compound produce 466 g of BaSO₄.

Ans. (40)

Sol. $S \rightarrow BaSO_4$

(atoms of S)_S = (atoms of S)_{BaSO₄}

$$\left(\frac{wt}{32}\right) \times N_A \times 1 = \left(\frac{466}{233}\right) \times N_A \times 1$$

$$(wt)_S = \left(\frac{466 \times 32}{233}\right)$$

$$(wt)_S = 64 \text{ g}$$

$$\% S = \frac{(wt)_S}{(wt)_{org. \text{ compound}}} \times 100 = \frac{64}{160} \times 100 = 40 \%$$

12. Find the spectral line of H-atom, which have $\lambda = 900 \text{ nm}$, $R_H = 10^5 \text{ cm}^{-1}$

(1) $n_2 = \infty \rightarrow n_1 = 1$, Lyman

(2) $n_2 = \infty \rightarrow n_1 = 2$, Balmer

(3) $n_2 = 5 \rightarrow n_1 = 3$, Paschan

(4) $n_2 = \infty \rightarrow n_1 = 3$, Paschan

Ans. (4)

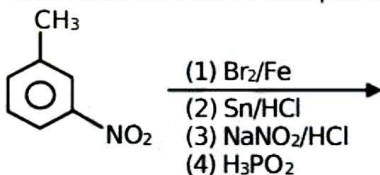
Sol. $\frac{1}{\lambda} = R_H \times 2^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$$\frac{1}{\lambda} = 10^5 \times 1 \times \left(\frac{1}{3^2} - \frac{1}{\infty^2} \right)$$

$$\frac{1}{\lambda} = \frac{10^5}{9}$$

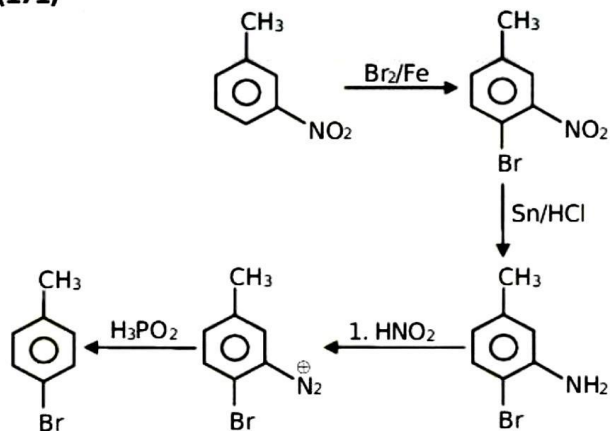
$$\Rightarrow \lambda = 9 \times 10^{-5} \text{ cm} = 900 \times 10^{-7} \text{ cm} = 900 \text{ nm}$$

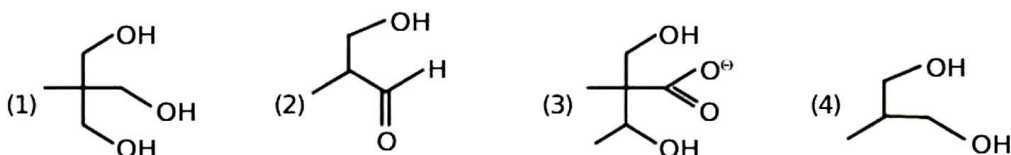
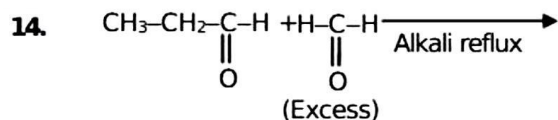
13. Find molecular mass of final product.



Ans. (171)

Sol.

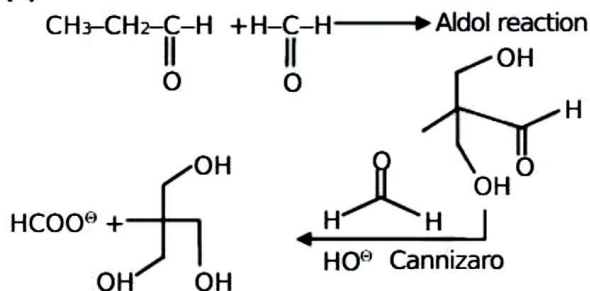




Ans.

(1)

Sol.



15. **Statement 1 :** Fructose gives silver mirror with Tollens reagent although --C(=O)H group is absent in it.

Statement 2 : Fructose in Alkaline (Base) medium converts into Aldose Sugar Glucose which has --C(=O)H group.

- (1) Both Statement 1 and statement 2 are true (2) Both statement 1 and statement 2 are false
 (3) Statement 1 is true but statement 2 is false (4) Statement 1 is false but statement 2 is true

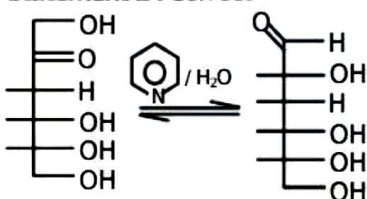
Ans.

(1)

Sol.

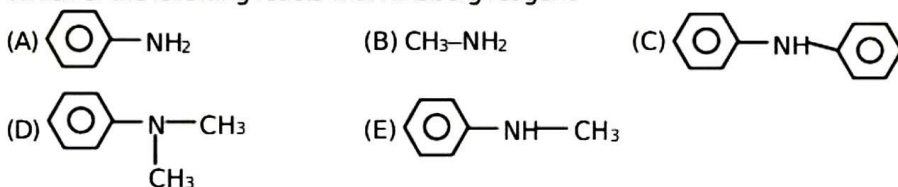
Statement 1 : Correct

Statement 2 : Correct



Rearrangement or inter conversion between fructose and glucose.

16. Which of the following reacts with Hinsberg reagent.



(1) A, B, C, E

(2) B, C, D

(3) A, C, D, E

(4) C, D, E

Ans.

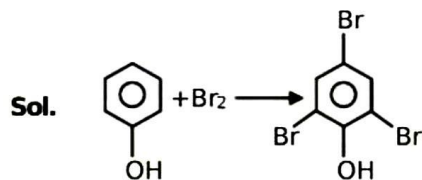
(1)

Sol.

Only primary and sec. amine reacts with Hinsberg reagent.

17. 2 g phenol react with Br₂ water to give trisubstituted phenol.
How much Br₂ is needed to complete reaction in grams. (Rounded off to nearest integer)

Ans. (10)

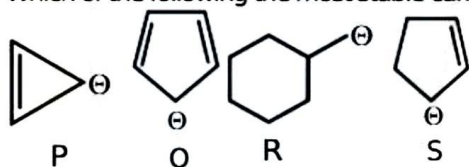


1 : 3 mol ratio

$$\frac{2\text{g}}{94} \quad \frac{2}{94} \times 3 \text{ mol Br}_2$$

Hence $\frac{2 \times 3}{94} \times 160\text{g Br}_2 \text{ used in reaction.} = 10.21 \text{ g}$

- 18 Which of the following the most stable carbanion is



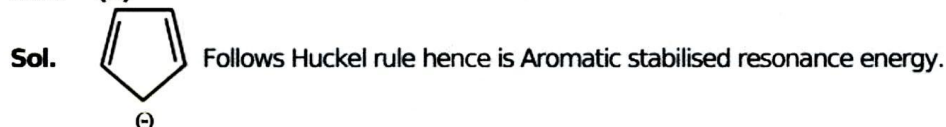
(1) P

(2) Q

(3) R

(4) S

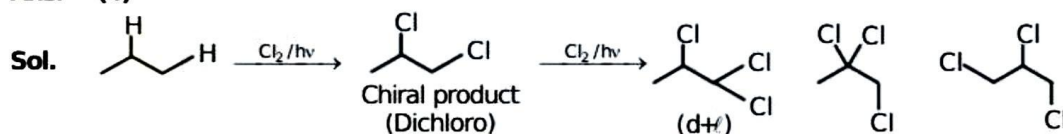
Ans. (2)



19. Propane reacts with Cl₂ in sunlight to give chiral product x which is dichloro product.

x is further chlorinated in sunlight to give how many trichloro product.

Ans. (4)



Total 4 product are formed.

20. In estimation of sulphur by carius method, 160 g of organic compound gives 466 g of Barium sulphate.
% of sulphur in the organic compound is.

Ans. (40)

Sol. BaSO₄(233)

$$\text{Moles} = \frac{466}{233} = 2$$

$$\text{Mass(s)} = 2 \times 32 = 64$$

$$\% \text{ of S} = \frac{64}{160} \times 100 = 40\%$$

21. Match the reactions name in given column to correct product formed.

Column-I		Column-II	
(A)	Wurtz fittig reaction	(P)	Fluoride product
(B)	Finkelstein Reaction	(Q)	Iodide product
(C)	Sand Meyer Reaction	(R)	Chloride product
(D)	Swart Reaction	(S)	Hydrocarbon product
(1) A-(S); (B)-(Q); (C)-(R); (D)-(P)		(2) A-(R); (B)-(P); (C)-(S); (D)-(P)	
(3) A-(P); (B)-(R); (C)-(Q); (D)-(R)		(4) A-(Q); (B)-(R); (C)-(S); (D)-(P)	

Ans. (1)